

RENSSELAER MECHATRONICS

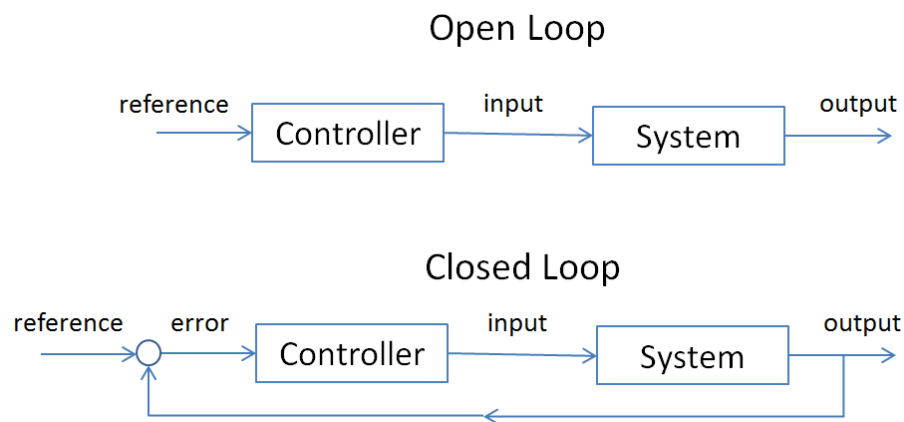
Basic DC Motor Position Control

Objectives:

- Control the position of the motor shaft
- Introduce basic control concepts
 - Closed loop feedback control
 - Instability: delays and controller gain

Background Information:

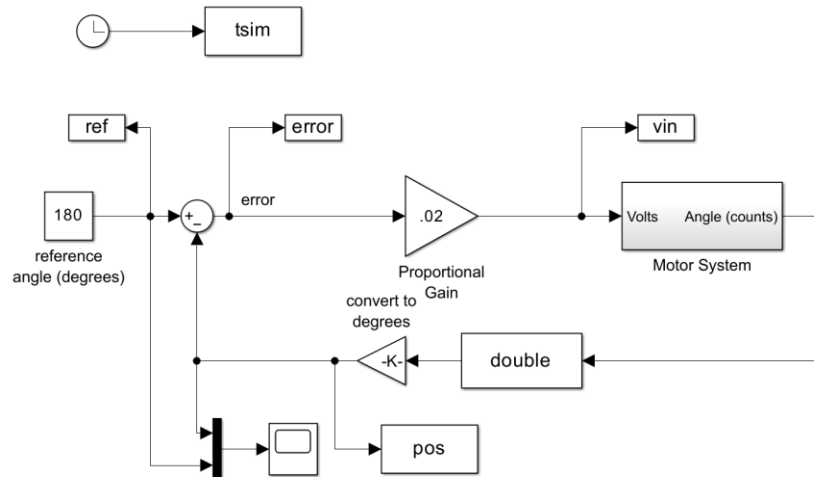
Controlling the output without directly measuring the output is called Open Loop control. If the output is measured and this information is used in the control this is called Closed Loop feedback control:



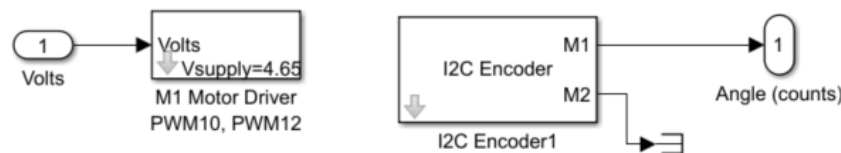
Measuring the output of interest and using this information in the control design is called closed loop feedback control. The basic strategy is to measure the output of interest, compare this to the desired output to generate an error signal and then try to reduce this error.

Proportional Control:

Build the following Simulink diagram:

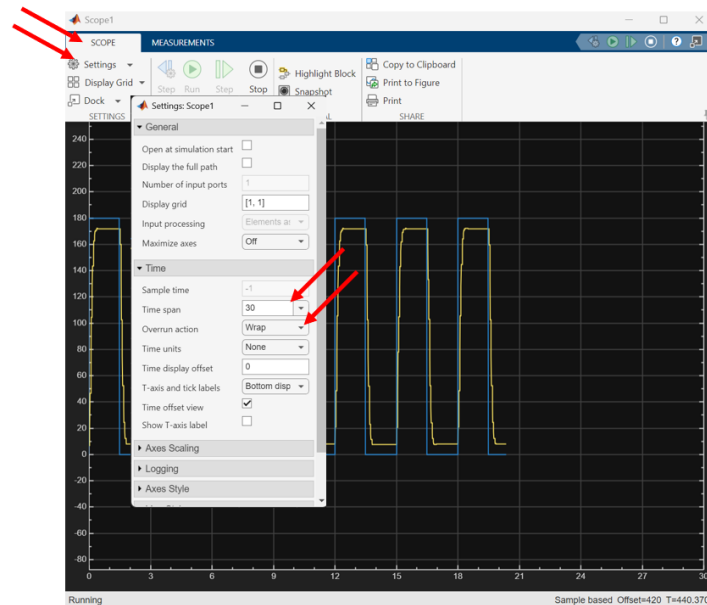


Tsim, vin, error, ref and pos are To Workspace blocks. The motor system is a subsystem created from the following blocks (review how to make a subsystem from previous labs. Use appropriate driver and encoder blocks for your system):



- **Make sure your battery switches are set to 'ON', 'VIN, and 'OFF' for the Driver Enable, Driver Voltage, and the Power Switch.**
- What value should be used to convert from the output in counts to degrees?
 - Verify this value. While running the code set the proportional gain to 0 (this turns off the motor). Turn the output motor shaft 1 revolution – do you get 360 degrees? If not calculate the correct gain and try again until you are SURE it is correct.
- If the position is initially zero and the control law is $V = K_p * (\text{reference} - 0)$, what is the maximum gain before the controller will saturate (demand more voltage than is available) when the reference is 180 degrees?
- Run the Simulink diagram. **If the error does not decrease initially the measured angle may need to have the sign changed (since the motor is going in the wrong direction). This can be done by adding a negative in the unit conversion block.**
 - Use a time step size of .02 seconds in external mode
 - Since the initial gain is zero the system should remain at rest
 - Change the proportional gain to .01 then change the reference to 180 then back to zero
 - Change the proportional gain to .025 then change the reference to 180 then back to zero
 - What value of the gain gives a response that has the least error?
 - What do you notice about the position of the motor shaft – does it move 180 degrees?

At this point, if you have observed the motor behavior as you change the gain. It will be easier to replace the constant reference with a pulse generator block. Change the constant block to a pulse generator block with amplitude 180, period of 4 seconds and a pulse width of 50%. Keep the simulation time infinite ('inf') and change the scope settings to "scroll" instead of "wrap" and use 30 seconds for the x axis in the setting so it is easier to observe the changes.



While this code is running change the gain from .01 to .025 and verify the same behavior with the constant block.

Modify simulation settings

- Change the time step size to 0.2 seconds and run in monitor and tune
- Change the proportional gain from .01 to .025. What do you notice?

Questions:

For this section **use a pulse generator block** with amplitude 180, period of 4 seconds and a pulse width of 50%. Use a stop time of 12 seconds – this will give you a couple step responses. Save the data for each test so you can easily plot a summary of your tests.

- Create a single plot that shows the reference and position vs. time for the following cases (you may want to use the 'save' command to save your data between each experiment for plotting later)
 - Step size of 0.02:
 - proportional gain of 0.01
 - proportional gain of 0.025
- Create a single plot that shows the reference and position vs. time for the following cases

- Step size of 0.2:
 - proportional gain of 0.01
 - Proportional gain of 0.025

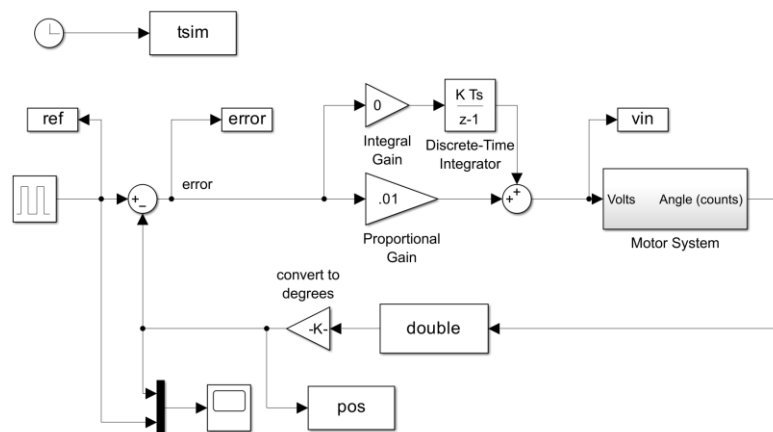
Comment on the effect of the varying the gain on the control system performance (steady state error, speed, stability). Does it get to 180?

Comment on the effect of the varying the step size on the control system performance (steady state error, speed, stability). How does the step size affect the performance?

Proportional Plus Integral Control:

Proportional control does not eliminate the steady state error. If the error signal is integrated this value will keep increasing as long as there is a steady state error, and a control effort can be applied that will reduce this error.

Build the following Simulink diagram:



- Run this on the system.
 - Use a time step size of 0.02
 - Change the pulse period to 30 seconds (this gives you a long time to see the results), change the display on the scope to display 60 seconds
 - Change the proportional gain to .005, integral gain to zero, – the response should have a noticeable steady state error (the error may be much larger in one direction)
 - About what amount is max steady state for a couple of pulses?
 - Change the integral gain to .005. What do you observe? What is the max steady state error now? Why can't you seem to get rid of the error?

Questions:

- Provide a plot showing the reference and position vs. time for a step size of 0.02, a proportional gain of .005 and a integral gain of .005. Show a couple of pulses that demonstrate the observed behavior.
- Provide a plot showing the reference and position vs. time for a step size of 0.02, a proportional gain of .025 and a integral gain of .025. Show a couple of pulses that demonstrate the observed behavior.